



HWA CHONG INSTITUTION  
2018 JC2 PRELIMINARY EXAMINATION

**MATHEMATICS**  
**Higher 2**

**9758/01**

Paper 1

**Tuesday**

**11 September 2018**

**3 hours**

Additional materials:      Answer paper  
                                    List of Formula (MF26)

**READ THESE INSTRUCTIONS FIRST**

Write your name and CT class on all the work you hand in, **including the Cover Page which is found on Page 2.**

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not write anything on the List of Formula (MF26).

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question. At the end of the examination, place the completed cover page on top of your answer scripts and fasten all your work securely together with the string provided.

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**This question paper consists of 8 printed pages.**



**HWA CHONG INSTITUTION**  
**2018 JC2 PRELIMINARY EXAMINATION**

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| CT                      | : | <table border="1"> <tr> <td>1</td> <td>7</td> <td></td> <td></td> <td></td> </tr> </table> |  |  |  |  | 1 | 7 |  |  |   |                                                                                          |  |  |  |  |  |  |  |  |
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| CENTRE / INDEX NO       | : | <table border="1"> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>   |  |  |  |  |   |   |  |  | / | <table border="1"> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> |  |  |  |  |  |  |  |  |
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### **Instructions to Candidates**

- 1 **Write** the information required in the box **above** and **calculator model(s)** used in the boxes **below**.
- 2 **Write the Question Number (in numerical order)** attempted in the left hand margin of the answer paper and in the left hand column of the box on this cover page.
- 3 **Arrange your answers in numerical order.**
- 4 \* Place this **cover page on top of your answer sheets.**  
\* Use the string provided **to tie this cover on top of your answer sheets securely.**  
\* Do not tie the answer sheets so tightly that the examiners cannot easily turn the pages.

**Graphing Calculator Model :**

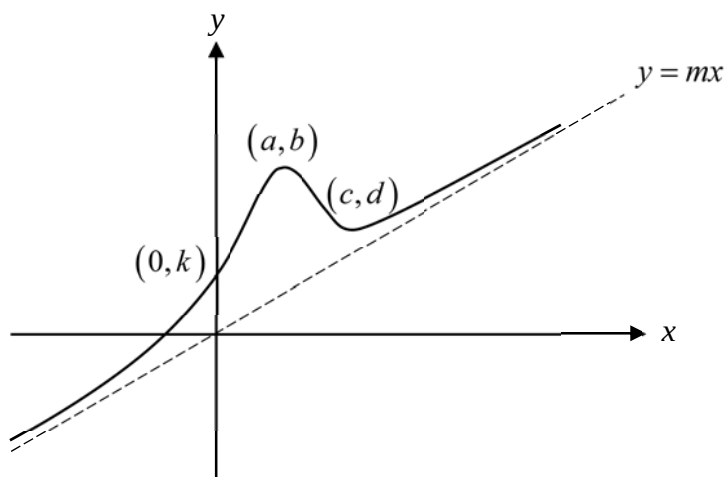
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**Scientific Calculator Model :**

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| For Candidate's Use | For Examiner's Use |         |
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- 1 The diagram below shows the graph of  $y = f(x)$ . The curve has a maximum point at  $(a, b)$ , a minimum point at  $(c, d)$  and cuts the  $y$ -axis at  $(0, k)$ . The equation of the asymptote is  $y = mx$ , where  $m$  is a positive constant.



Sketch the graph of  $y = f'(x)$ , giving equation(s) of any asymptotes and coordinate(s) of any intercepts, if it is possible to do so. [2]

- 2 A drone is programmed to fly from a point  $O$  at ground level eastwards to a point  $C$  at the top of a building. The point  $C$  is 0.1 km vertically above ground level and 0.6 km horizontally from  $O$ . The drone passes through two checkpoints  $A$  and  $B$  before reaching  $C$ . The horizontal distances and vertical heights of  $A$  and  $B$  are shown below.

| Point | Horizontal Distance from $O$ (km) | Vertical Height (km) |
|-------|-----------------------------------|----------------------|
| $A$   | 0.15                              | 0.1                  |
| $B$   | 0.3                               | 0.125                |

It is given that the flight path of the drone is cubic in nature. Taking  $O$  to be the origin and  $O, A, B$  and  $C$  all lie on the same vertical plane, find the cartesian equation of the flight path. [3]

The drone manufacturer is highly confident of the ability of the drone to keep to its flight path despite external factors such as wind, temperature or humidity. It claims that

“The drone on this particular programmed flight path will pass through the point at a vertical height of 0.1125 km and a horizontal distance of 0.45 km eastwards from  $O$ .”

By considering the flight path of the drone, comment on the accuracy of the manufacturer's claim. [1]

- 3 (a) A point  $Q$  has position vector  $\mathbf{q}$  and a line  $l$  has equation  $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$ ,  $\lambda \in \mathbb{R}$ . A vector  $\mathbf{p}$ , where  $|\mathbf{p}| = 1$ , is in the direction of  $\mathbf{d}$ . Give a geometrical meaning of  $|\mathbf{p} \times (\mathbf{a} - \mathbf{q})|$ . [1]

- (b) The vectors  $\mathbf{a}$  and  $\mathbf{b}$  form two adjacent sides of a triangle  $OAB$ , where  $\mathbf{a} = \overrightarrow{OA}$  and  $\mathbf{b} = \overrightarrow{OB}$ . The angle between  $\mathbf{a}$  and  $\mathbf{b}$  is  $\theta$ . By considering  $(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$  and the cosine rule for the triangle  $OAB$ , show that  $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$ . [4]

- 4 (i) Show that  $\frac{3}{2} - \sqrt{a + \frac{1}{4}} < 1$ , where  $a$  is a positive constant. [1]

- (ii) Solve the inequality  $\frac{x^2 - x - a}{x - 1} \leq 2$ , where  $a$  is a positive constant. [4]

- (iii) Hence solve  $\frac{a + x - x^2}{x} \leq 2$ . [2]

- 5 Let  $f(r) = \cos \left[ \alpha + \left( r + \frac{1}{2} \right) \beta \right]$ ,  $\beta \neq 2k\pi$ ,  $k \in \mathbb{Z}$ .

- (i) Find  $f(r) - f(r-1)$ . [2]

- (ii) Show that

$$\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + n\beta) = \frac{\sin(\alpha + p\beta) \sin q\beta}{\sin \frac{1}{2}\beta},$$

where  $p$  and  $q$  are constants to be determined in terms of  $n$ . [4]

- (iii) Deduce an expression for

$$\cos \alpha + \cos \left( \alpha + \frac{\pi}{2} \right) + \cos(\alpha + \pi) + \cos \left( \alpha + \frac{3\pi}{2} \right) + \dots + \cos \left( \alpha + \frac{n\pi}{2} \right). \quad [2]$$

- 6 A curve  $C$  has parametric equations

$$x = 2 \sin t, \quad y = 1 + \cos t, \quad 0 < t < \pi.$$

- (i) Show that the equation of the tangent to  $C$  at the point  $P(2 \sin p, 1 + \cos p)$  is  $2y + x \tan p = 2(1 + \sec p)$ . [4]

- (ii) The tangent at  $P$  meets the  $x$ -axis at the point  $A$  and the  $y$ -axis at the point  $B$ . The point  $M$  is the midpoint of  $AB$ . Find the cartesian equation of the curve traced by  $M$  as  $p$  varies. [5]

- 7 Mr Wang decides to paint the walls of his room which has a total surface area of  $40 \text{ m}^2$ . He has two plans:

Plan A: He paints  $7 \text{ m}^2$  on the first day, and on each subsequent day, he paints  $0.5 \text{ m}^2$  less than the previous day.

Plan B: He paints  $7 \text{ m}^2$  on the first day, and on each subsequent day, he paints 20% less than the previous day.

- (i) Find an expression for the area painted on the  $n$ th day according to Plan A. Give your answer in terms of  $n$ . [1]
- (ii) Find the number of days required for him to complete painting his room according to Plan A. [2]
- (iii) Explain why Plan A cannot be used to paint a wall of arbitrary size. Find the largest area that Plan A can be applied to. [2]
- (iv) Find algebraically the number of days needed to paint at least 70% of his room according to Plan B. [3]
- (v) State, with a reason, which plan A or B Mr Wang should choose to complete the paint job. [1]

- 8 The curve  $G$  has equation

$$y = \frac{x+2}{x(x+k)},$$

where  $k$  is a real constant.

- (i) Find the range of values of  $k$  for which  $G$  has no stationary points. [3]

In the rest of the parts of the question, let  $k = -2$ .

- (ii) Sketch  $G$ , stating clearly the equations of any asymptotes, the coordinates of the stationary points and the points where  $G$  crosses the axes. [3]
- (iii) State the values of  $m$  for which the line  $y = m$  intersects  $G$  once. [1]
- (iv) By sketching a suitable curve on the diagram in part (ii), show that the equation  $x^4 - 4x^3 + 3x^2 + x - 2 = 0$  has exactly two real roots. [2]

- 9 (a) Showing your working clearly, find the complex numbers  $z$  and  $w$  which satisfy the simultaneous equations

$$\begin{aligned} 4iz - w &= 9i - 13, \\ (4 + 2i)w^* &= z + 3i. \end{aligned} \quad [4]$$

- (b) The complex numbers  $u$  and  $v$  are such that  $u = 5e^{\frac{7}{12}\pi i}$  and  $v = 6ie^{-\frac{1}{3}\pi i}$  respectively.

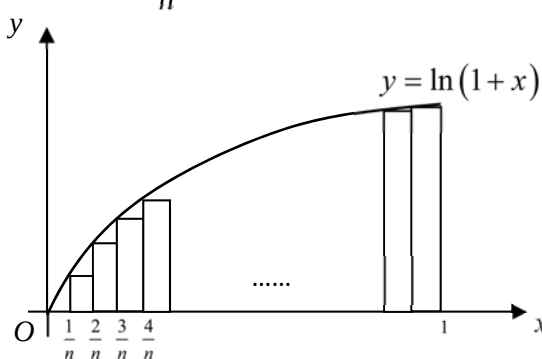
- (i) Find an exact expression of  $\frac{u^2}{v^*}$ , giving your answer in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . [4]

- (ii) Find the three smallest positive integer values of  $n$  for which  $\frac{u^n}{v^*}$  is purely imaginary. [2]

- 10 (a) (i) Find  $\int x\sqrt{x-1} \, dx$  using integration by parts. [2]

- (ii) The shape of a metal sculpture is formed by rotating the region bounded by the curve  $y = \sqrt{a + x\sqrt{x-1}}$ , where  $a$  is a positive integer, the lines  $x=1$  and  $y = \sqrt{a+30}$ , through  $2\pi$  radians about the  $x$ -axis. Find the exact volume of the metal sculpture, giving your answer in terms of  $\pi$ . [4]

- (b) (i) The diagram below shows a sketch of the graph of  $y = \ln(1+x)$  for  $0 \leq x \leq 1$ . Rectangles each of width  $\frac{1}{n}$  are drawn under the curve for  $0 \leq x \leq 1$ .

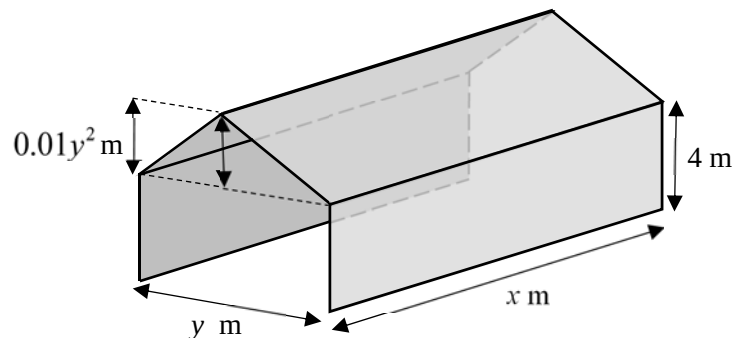


Show that  $A$ , the total area of all the rectangles, is given by

$$A = \frac{1}{n} \ln \left[ \frac{(n+1)(n+2)(n+3) \dots (2n-1)}{n^{n-1}} \right]. \quad [2]$$

- (ii) Find the exact value of  $\lim_{n \rightarrow \infty} A$ . [3]

- 11** To celebrate Singapore's Bicentennial in 2019, the organising committee plans to hold a banquet at the Padang. A tent for the banquet consisting of two rectangular vertical sides and two pieces of the roof is to be constructed as shown in the diagram below. The tent has a length of  $x$  m and width  $y$  m, and a total floor area of  $4000 \text{ m}^2$ . The vertical sides of the tent are 4 m tall, and the roof adds another  $0.01y^2$  m to the overall height of the tent.



- (i)** Show that  $A$ , the total external surface area of the tent is given by

$$A = 8x + 4000\sqrt{\frac{6400}{x^2} + 1}. \quad [3]$$

- (ii)** Show that if  $A$  has a stationary value for some  $x$ , then  $x$  satisfies the equation

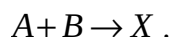
$$\frac{a}{x^6} = \frac{6400}{x^2} + 1, \text{ where } a \text{ is a constant to be determined.} \quad [3]$$

- (iii)** If the material for the tent costs \$2.50 per  $\text{m}^2$ , estimate the minimum total cost of the material for the whole tent. [3]

- (iv)** A scale model of the tent is to be 3D-printed in a process where the total external surface area of this model always satisfies the equation  $A = 8x + 4000\sqrt{\frac{6400}{x^2} + 1}$ .

If  $x$  increases at a rate of 2 units per minute, use differentiation to find the rate of change in the total external surface area at the instant when  $x = 200$  units. [3]

- 12** An experiment is conducted at room temperature where two substances,  $A$  and  $B$ , react in a chemical reaction to form  $X$  as shown below:



The initial concentrations in  $\text{mol/dm}^3$  of substances  $A$  and  $B$  are  $a$  and  $b$  respectively. At time  $t$  seconds, the concentration of  $A$  and  $B$  are each reduced by  $x$ , where  $x$  denotes the concentration of  $X$  at time  $t$ .

- (i)** State the concentrations of  $A$  and  $B$  at time  $t$ . [1]
- (ii)** It is known that the rate of change of concentration of  $X$  at time  $t$  is proportional to the product of concentration of  $A$  and  $B$  at time  $t$  with a constant of proportionality  $k$ . Write down a differential equation involving  $x$ ,  $a$ ,  $b$ ,  $t$  and  $k$ . [1]
- (iii)** State the maximum value of  $x$  if  $a \leq b$ . Justify your answer. [2]

In the rest of the parts of the question, assume  $a = b$ .

- (iv)** The initial concentration of  $X$  is zero. Solve the differential equation in part **(ii)**, leave your answer in terms of  $x$ ,  $a$  and  $t$ .

Express the solution in the form  $x = f(t)$  and sketch  $x = f(t)$  relevant in this context. Label the graph as  $S_1$ . [5]

It is known that the rate of change of concentration of  $X$  is doubled with every  $10^\circ\text{C}$  rise in the temperature. The experiment above is repeated but at a temperature  $20^\circ\text{C}$  above the room temperature. The concentration of  $X$  for this 2nd experiment at time  $t$  is now denoted by  $x_2$ . Let  $S_2$  be the solution curve for the 2nd experiment.

- (v)** State an equation relating the rate of change of concentration of  $x$  and the rate of change of concentration of  $x_2$  at time  $t$ . [1]

There is no need to solve for  $S_2$  for the rest of the parts of the question.

- (vi)** On the same diagram as in part **(iv)**, sketch the solution curve for  $S_2$ . Show clearly the relative positions of  $S_1$  and  $S_2$  and their behaviour when  $t \rightarrow \infty$ . [2]
- (vii)** It is given that  $S_1$  passes through the point  $(1, \alpha)$  and  $S_2$  passes through the point  $(1, \beta)$ . Using the rate of change of concentration of  $X$  for the two experiments, state the inequality relating  $\alpha$  and  $\beta$  in this context, justifying your answer. [2]